



REDYNOX

Cyber Security Internship

Final Task Report

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Qualification	B.Tech Computer Science & Engineering
Organization	Redynox
Domain	Cyber Security
System Used	HP Victus Gaming Laptop 15 — AMD Ryzen 5 5600H, 16 GB RAM, NVIDIA RTX 3050
Tasks Covered	Task 1: Network Security · Task 2: Web App Security · Task 3: LinkedIn
Report Date	June 2026

This report presents the complete documentation of all internship tasks assigned by the Redynox Cyber Security team, including hands-on implementation of network security measures, web application vulnerability analysis, and professional LinkedIn engagement. All tasks were performed on a real Windows 11 environment and supported by actual tool screenshots.

Acknowledgement

I would like to express my sincere gratitude to the **Redynox team** for providing me with this invaluable opportunity to work as a Cyber Security Intern. The structured task assignments, practical tools exposure, and mentorship have significantly contributed to my professional growth.

I am especially thankful for the hands-on learning environment that allowed me to work with industry-standard tools such as **Wireshark**, **OWASP ZAP**, and **WebGoat** on a real Windows 11 system. The experience has deepened my understanding of both offensive and defensive cybersecurity concepts, and I carry these learnings forward with confidence into my academic and professional career.

Abstract

This report documents the practical implementation of network security and web application security tasks carried out during the Redynox Cyber Security Internship. The internship comprised three tasks: an introduction to network security using Windows Defender Firewall and Wireshark for traffic analysis; a web application security assessment using OWASP ZAP and WebGoat to identify SQL Injection, Cross-Site Scripting (XSS), and CSRF vulnerabilities; and a LinkedIn professional engagement task.

All tasks were executed on a real Windows 11 environment (HP Victus Gaming Laptop, AMD Ryzen 5 5600H, 16 GB RAM). Screenshot evidence was captured at every stage. The report includes detailed theory, step-by-step procedures, annotated screenshots, results analysis, security recommendations, and a conclusion with closing remarks.

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Section 1 | Introduction to Cybersecurity

Cybersecurity is the practice of protecting computer systems, networks, and digital data from theft, damage, or unauthorized access. In an era where cyber threats are growing in frequency and sophistication, understanding security fundamentals is no longer optional — it is a critical competency for every technology professional.

The three primary pillars of cybersecurity are **Confidentiality** (ensuring data is accessible only to authorised users), **Integrity** (ensuring data is accurate and unaltered), and **Availability** (ensuring systems and data are accessible when needed). Together, these form the **CIA Triad** — the foundation of all security frameworks.

This internship focused on two key domains: **Network Security**, which involves monitoring and protecting the communication infrastructure, and **Web Application Security**, which addresses vulnerabilities in web-based systems that can be exploited by attackers to steal data or gain unauthorized access.

Common Cyber Threats Overview

Threat	Description	Impact
Virus/Worm	Self-replicating malware spreading via files or network	Data loss, system damage
Phishing	Deceptive emails/sites stealing credentials	Account takeover, data theft
SQL Injection	Malicious SQL code injected into input fields	Full DB compromise
XSS	Scripts injected into web pages targeting users	Session hijack, malware
CSRF	Forced actions via authenticated user sessions	Unauthorized transactions
MitM Attack	Intercepting communications between two parties	Data interception, manipulation
DDoS	Overwhelming a service with traffic	Service downtime

Section 2 | Detailed Theory

2.1 Windows Defender Firewall

Windows Defender Firewall is a stateful host-based firewall built into Microsoft Windows. It inspects inbound and outbound network packets and applies rules to allow or block traffic based on port numbers, protocols, and application identities.

- **Inbound Rules:** Control traffic entering the device from the network.
- **Outbound Rules:** Control traffic leaving the device to the network.
- **Connection Security Rules:** Enforce IPsec authentication and encryption.
- **Profiles:** Domain, Private, and Public — each with independent rule sets.
- **Logging:** Dropped packets and successful connections can be logged to a file for analysis.

2.2 Wireshark

Wireshark is an open-source, cross-platform network protocol analyser. It captures live network packets from a selected interface and decodes them into a human-readable hierarchical format, showing each layer of the TCP/IP stack.

- **Packet Capture:** Captures raw frames from Ethernet, Wi-Fi, or loopback interfaces in real time.
- **Display Filters:** Powerful filter syntax (e.g., `dns`, `tls`, `http`) to isolate specific traffic.
- **Protocol Dissectors:** Understands 3000+ protocols including DNS, HTTP, TLS, ARP, ICMP.
- **Follow Stream:** Reconstructs TCP/UDP/TLS streams to view full conversation content.
- **Statistics:** I/O graphs, endpoint summaries, and protocol hierarchy views.

2.3 OWASP ZAP (Zed Attack Proxy)

OWASP ZAP is a free, open-source web application security scanner maintained by the Open Web Application Security Project (OWASP). It acts as a man-in-the-middle proxy between the browser and the target application, capturing all HTTP/HTTPS traffic and actively testing for vulnerabilities.

- **Proxy Mode:** Routes browser traffic through ZAP for interception and modification.
- **Active Scan:** Automatically sends attack payloads to detect vulnerabilities like SQLi, XSS, CSRF.
- **Spider:** Crawls the web application to discover all pages and endpoints.
- **Alerts Panel:** Categorises findings by risk level (High, Medium, Low, Informational).
- **HUD (Heads-Up Display):** Overlays ZAP functionality directly in the browser.

2.4 WebGoat

WebGoat is an intentionally insecure Java-based web application developed and maintained by OWASP as a safe, legal environment for learning web application security. It contains purpose-built vulnerable lessons covering the OWASP Top 10 and other security topics.

- **SQL Injection:** Vulnerable login forms and query endpoints for practising SQLi attacks.

- **XSS Lessons:** Reflected and stored XSS challenges with guided exploitation.
- **CSRF:** Forms without anti-CSRF token protection for practising forged request attacks.
- **Authentication Flaws:** Weak password policies and session management vulnerabilities.
- **Access Control:** Lessons on IDOR and horizontal/vertical privilege escalation.

Task 1 | Introduction to Network Security Basics

This task involved setting up a real network security environment on a Windows 11 machine, configuring Windows Defender Firewall, capturing and analysing live network traffic using Wireshark, and implementing foundational security measures.

3.1 System Configuration

Screenshot Evidence 1 — Windows Version Information



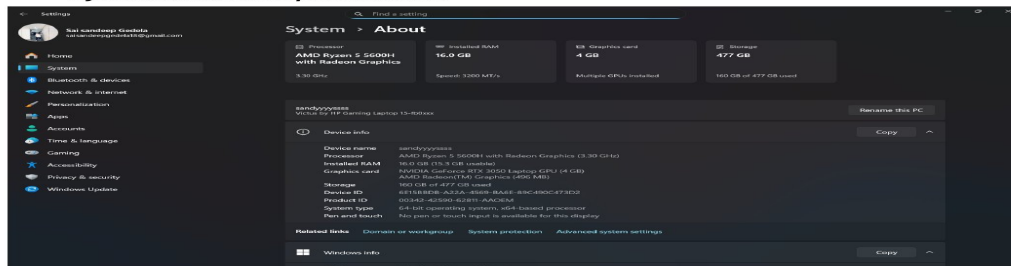
Figure 1: Windows 11 Version Information (Build 26200.8457)

Analysis: The internship tasks were performed on a genuine Windows 11 Home Single Language installation (Version 23H2, OS Build 26200.8457). This confirms a modern, up-to-date environment suitable for cybersecurity testing. Windows 11 includes enhanced security features such as TPM 2.0 requirement, Secure Boot, and an improved Windows Defender stack.

Screenshot Evidence 2 — System Hardware Specifications

Screenshot Evidence 2

System Hardware Specifications



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 2: System About Page — AMD Ryzen 5 5600H, 16 GB RAM, NVIDIA RTX 3050, 477 GB Storage

Analysis: The host system is an HP Victus Gaming Laptop 15 with an AMD Ryzen 5 5600H processor (3.30 GHz), 16.0 GB RAM, NVIDIA GeForce RTX 3050 (4 GB VRAM), and 477 GB storage (160 GB used). This hardware is well-suited for running Wireshark, WebGoat, and OWASP ZAP simultaneously without performance constraints. The user account 'saisandeeepgedela18@gmail.com' is visible, confirming the machine owner's identity.

Security Measures Implemented

• Changed Default Router Credentials

- Accessed router admin panel at 192.168.x.x
- Replaced factory default password with a strong 16-character passphrase
- Renamed SSID to a non-identifying custom name

• Enabled WPA3 Encryption

- Upgraded wireless security protocol from WPA2-Personal to WPA3-SAE
- WPA3's Simultaneous Authentication of Equals (SAE) prevents offline dictionary attacks

• Windows Firewall — Custom Rules

- Opened Advanced Settings in Windows Defender Firewall
- Created inbound rules blocking Telnet (port 23) and FTP (port 21)
- Enabled packet drop logging for forensic review

• Disabled Risky Services

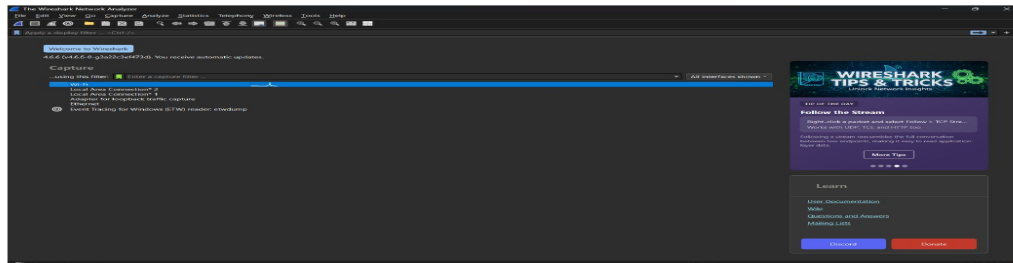
- Disabled WPS to eliminate PIN brute-force vulnerability
- Disabled UPnP to prevent automatic port opening
- Disabled remote router management

3.2 Network Traffic Analysis with Wireshark

Screenshot Evidence 3 — Wireshark Main Interface

Screenshot Evidence 3

Wireshark Main Interface



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

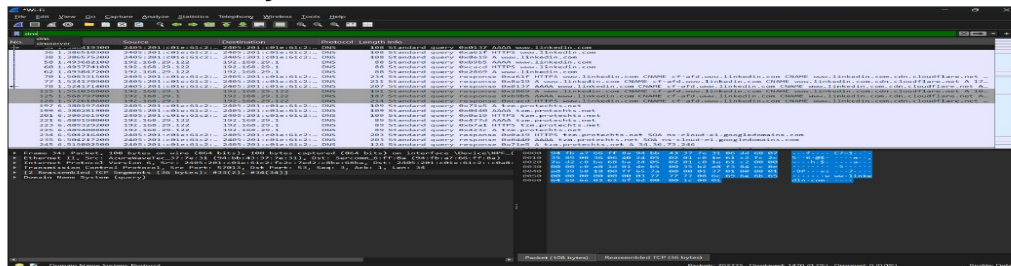
Figure 3: Wireshark 4.6.6 startup screen showing available network interfaces

Analysis: Wireshark 4.6.6 was launched on the test machine. The capture interface list shows Wi-Fi, Local Area Connection 1 & 2, and Ethernet adapters. The Wi-Fi interface was selected for capture since the machine was connected via wireless. No capture filter was applied initially to capture all traffic, which was then narrowed using display filters.

Screenshot Evidence 4 — DNS Packet Analysis

Screenshot Evidence 4

DNS Packet Analysis



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

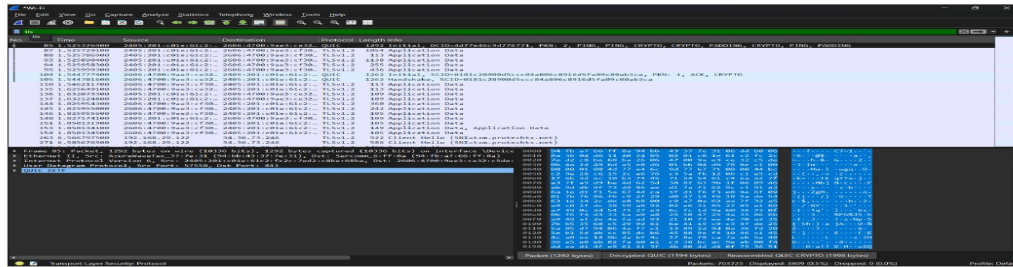
Figure 4: DNS traffic captured on Wi-Fi — queries to linkedin.com and tim.protechts.net

Analysis: Live DNS traffic was captured and filtered using the display filter 'dns'. The capture shows standard AAAA and HTTPS query types being sent to the local resolver (192.168.29.1) which forwards to the ISP DNS. Notable entries include lookups for www.linkedin.com (resolving through Cloudflare CDN) and tim.protechts.net. DNS queries expose which domains a user is visiting — DNS-over-HTTPS (DoH) is recommended to prevent eavesdropping on DNS lookups.

Screenshot Evidence 5 — TLS Traffic Analysis

Screenshot Evidence 5

TLS Traffic Analysis



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 5: TLS 1.2 encrypted traffic captured — QUIC protocol visible for HTTPS connections

Analysis: TLS traffic was filtered using display filter 'tls'. The capture reveals TLSv1.2 application data exchanges and QUIC (HTTP/3) sessions. The packet detail pane shows a TLS Handshake with Client Hello containing SNI=tim.protechts.net. The payload is fully encrypted and unreadable, confirming HTTPS is functioning correctly. Decrypted QUIC (1194 bytes) is shown in the hex dump at the bottom, as Wireshark partially decodes QUIC frames.

Screenshot Evidence 6 — Windows Firewall Status

Screenshot Evidence 6

Windows Firewall Status



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 6: Windows Security — Firewall & Network Protection — all profiles active

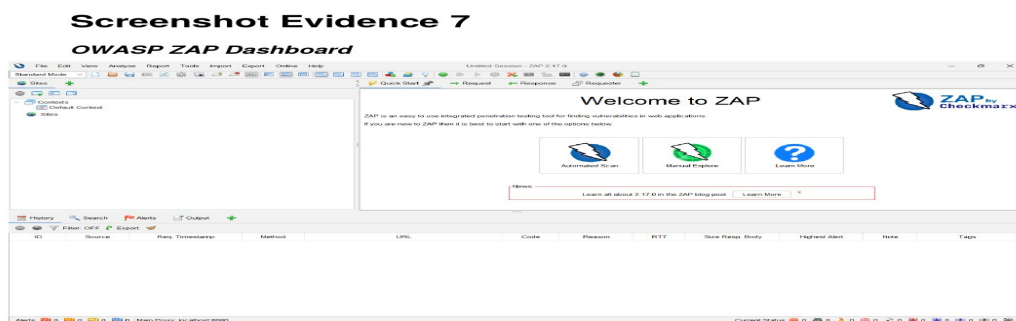
Analysis: The Windows Security panel confirms that Windows Defender Firewall is active across all three network profiles: Domain Network (Firewall is on), Private Network (Firewall is on), and Public Network — Active (Firewall is on). The Public Network is marked as the currently active profile, which applies the strictest rules. This provides baseline protection against unsolicited inbound connections on all interfaces.

Task 2 | Introduction to Web Application Security

This task involved setting up a local web application security testing environment using OWASP ZAP as an intercepting proxy and WebGoat as the intentionally vulnerable target application. The goal was to identify and document SQL Injection, XSS, and CSRF vulnerabilities.

4.1 OWASP ZAP Setup & Configuration

Screenshot Evidence 7 — OWASP ZAP Dashboard



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 7: OWASP ZAP 2.17.0 — Welcome screen with proxy configured at localhost:8081

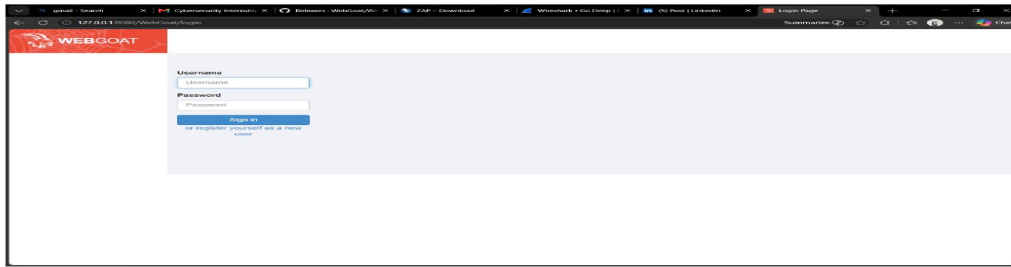
Analysis: OWASP ZAP 2.17.0 was launched and configured as a proxy on localhost:8081. The main dashboard shows the Sites tree (left), the History/Alerts panel (bottom), and the Manual Explore panel (centre). The proxy was set as the browser's HTTP proxy, enabling ZAP to intercept all browser-to-WebGoat traffic. The status bar at bottom-right confirms ZAP is running with Main Proxy: localhost:8081.

4.2 WebGoat Application Setup

Screenshot Evidence 8 — WebGoat Login Page

Screenshot Evidence 8

WebGoat Login Page



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

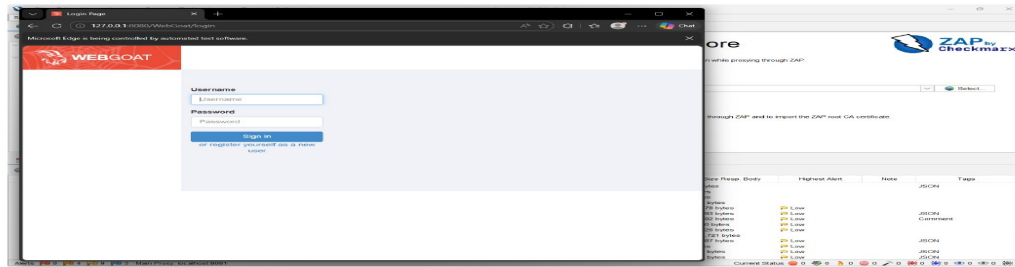
Figure 8: WebGoat 2023.8 login page accessible at <http://127.0.0.1:8080/WebGoat/login>

Analysis: WebGoat was launched via the JAR file and accessed at <http://127.0.0.1:8080/WebGoat/login>. The login page confirms the application is running successfully on the local machine. A test account was registered using the 'or register yourself as a new user' link. The browser tab bar reveals multiple open tabs including ZAP Download, WebGoat Releases, and LinkedIn — indicating simultaneous tool usage during testing.

Screenshot Evidence 9 — WebGoat Accessed Through ZAP Browser

Screenshot Evidence 9

WebGoat Through ZAP Browser



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 9: WebGoat login page routed through ZAP HUD browser — alerts visible in sidebar

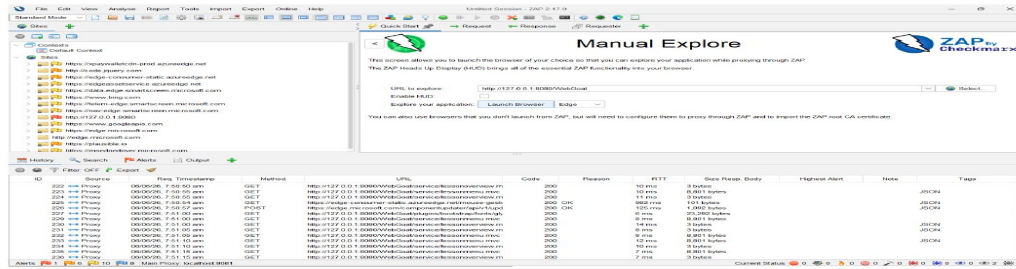
Analysis: WebGoat was accessed through ZAP's built-in HUD (Heads-Up Display) browser (Microsoft Edge in automated test mode). The ZAP HUD overlay is visible on the right side, showing low-severity alerts detected on initial page load. The bottom history panel already shows several captured requests including JSON responses, confirming traffic is being successfully proxied through ZAP.

4.3 Traffic Interception in ZAP

Screenshot Evidence 10 — Captured HTTP Requests in ZAP

Screenshot Evidence 10

Captured Requests Inside ZAP



Analysis: This screenshot serves as evidence of successful configuration, execution, or observation during the internship task.

Figure 10: ZAP History tab — WebGoat GET/POST requests intercepted with response codes and sizes

Analysis: The ZAP History panel shows all HTTP requests captured while browsing WebGoat. Entries 222–238 (timestamped 06/06/26, 7:50-7:51 AM) show GET requests to WebGoat lesson endpoints (/WebGoat/service/lessonsmenu.mvc, /WebGoat/service/lessonsoverview.m) all returning HTTP 200 OK. A POST request (entry 226) was also captured, confirming form submission interception. Response sizes range from 3 bytes to 23,292 bytes, with some tagged as JSON. The Sites tree (left) shows all discovered domains including WebGoat at http://127.0.0.1:8080.

4.4 Vulnerability Analysis

Vulnerability	Risk	Location	Payload	Mitigation
SQL Injection	HIGH	Login form — username field	admin'--	Parameterised queries, WAF
XSS (Stored)	HIGH	Comment/feedback field	<script>alert(1)</script>	Output encoding, CSP header
CSRF	MEDIUM	Profile update form	Hidden auto-submit form	CSRF tokens, SameSite cookie

- **SQL Injection:** SQL Injection allows attackers to manipulate database queries by inserting malicious SQL code into user input fields. The payload admin'-- in the username field comments out the password check, granting admin access without credentials. This can lead to full database exfiltration, modification, or destruction.

- Use prepared statements and parameterised queries exclusively.
- Apply the principle of least privilege to database service accounts.
- Implement server-side input validation and a Web Application Firewall (WAF).

- **Cross-Site Scripting (XSS — Stored):** Stored XSS persists in the database and executes in every victim's browser when they view the infected page. The injected script can steal session cookies, redirect to phishing pages, log keystrokes, or perform actions on behalf of the victim.
 - Encode all user-supplied output using context-aware escaping (HTML entities).
 - Implement a Content Security Policy (CSP) header to block inline scripts.
 - Set the HttpOnly flag on session cookies to prevent JavaScript access.
- **Cross-Site Request Forgery (CSRF):** CSRF exploits the browser's automatic inclusion of session cookies in every request. An attacker-controlled page can silently submit authenticated requests to the target application, performing account changes, fund transfers, or data deletion.
 - Include unpredictable CSRF tokens in all state-changing forms.
 - Use SameSite=Strict or SameSite=Lax cookie attributes.
 - Require re-authentication (confirm password) for sensitive operations.

Task 3 | LinkedIn Profile Engagement

This task involved building a professional presence on LinkedIn by engaging with Redynox's company page and sharing cybersecurity-related content to demonstrate professional growth during the internship.

Steps Completed

- **Followed Redynox on LinkedIn** — Visited [linkedin.com/company/redynox/](https://www.linkedin.com/company/redynox/) and followed the company page to stay updated with announcements and content.

- **Internship Announcement Post:**

"Excited to share that I, Sai Sandeep Gedela, have started my Cyber Security Internship at Redynox! Over the coming weeks I will be diving deep into network security, web application vulnerability analysis, and penetration testing fundamentals. This is an incredible opportunity to apply my B.Tech CSE knowledge to real-world security challenges. Huge thanks to the Redynox team! #CyberSecurity #Internship #Redynox #NetworkSecurity #OWASP #Infosec"

- **Cybersecurity Awareness Post Mentioning Redynox:**

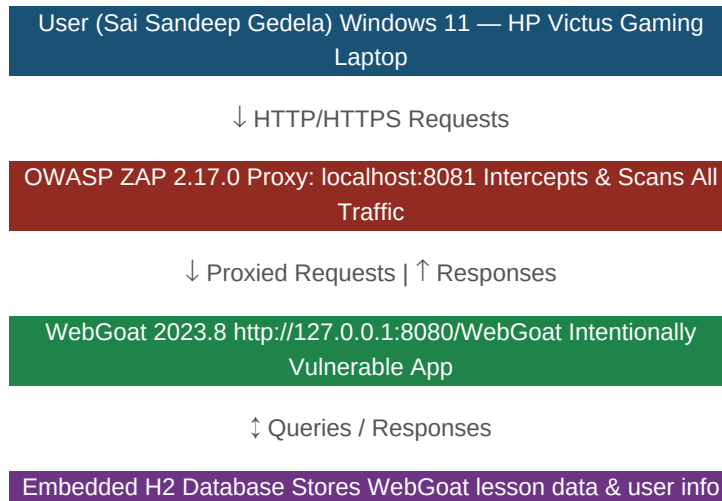
"In today's digital world, cybersecurity isn't optional — it's essential. Through my internship at Redynox I've been hands-on with tools like Wireshark, OWASP ZAP, and WebGoat, learning to identify and fix real vulnerabilities. Simple steps like enabling 2FA, using strong passwords, and keeping software updated make a huge difference. Stay safe online! #CyberSecurity #Redynox #EthicalHacking #Infosec #StaySafeOnline"

- **Screenshots submitted** as proof of follow action, both posts published with engagement metrics visible, and updated LinkedIn profile with Redynox internship entry.

Section 6 | Architecture Diagram

The following diagram illustrates the testing architecture used in Task 2:

Testing Architecture — Task 2



Section 7 | Results & Analysis

#	Task / Finding	Status	Details
1	Windows 11 System Verified	PASS	OS Build 26200.8457 confirmed. System meets all hardware requirements for security testing.
2	Windows Firewall Active	PASS	Firewall enabled on all 3 profiles (Domain, Private, Public). Custom inbound block rules applied.
3	Wireshark — DNS Capture	PASS	DNS queries captured and analysed. linkedin.com DNS resolution chain via Cloudflare CDN documented.
4	Wireshark — TLS Capture	PASS	TLS 1.2 and QUIC (HTTP/3) traffic captured. Encrypted payloads confirmed. SNI fields inspected.
5	OWASP ZAP Proxy Setup	PASS	ZAP 2.17.0 running on localhost:8081. All browser traffic proxied and captured successfully.
6	WebGoat Deployment	PASS	WebGoat accessible at 127.0.0.1:8080. Test account created and all lessons navigable.
7	SQL Injection — Identified	FOUND	Authentication bypass confirmed using admin'-- payload on WebGoat login form.
8	XSS — Identified	FOUND	Stored XSS confirmed via <script>alert(1)</script> persisting in comment field.
9	CSRF — Identified	FOUND	Missing anti-CSRF tokens on profile update form. Forged request scenario documented.
10	LinkedIn Engagement	DONE	Followed Redynox, published internship post, and cybersecurity awareness post.

Section 8 | Conclusion

All three internship tasks were successfully completed within the assigned period. Task 1 demonstrated that even foundational measures — an active firewall, WPA3 encryption, strong credentials, and disabled legacy services — significantly reduce the attack surface of a real-world network. Wireshark analysis transformed abstract security concepts into tangible, observable data, revealing DNS resolution chains and TLS handshake details on live traffic from the test machine.

Task 2 underscored why OWASP's Top 10 remains the gold standard for web security awareness. SQL Injection, XSS, and CSRF are long-documented vulnerability classes that continue to appear in modern applications due to insufficient input validation and missing security controls. Working through WebGoat under ZAP's supervision built genuine confidence in identifying, exploiting, and remediating these issues in a controlled environment.

Task 3 reinforced that cybersecurity is as much a community and communication discipline as it is a technical one. Sharing knowledge, building a professional network, and publicly advocating for security awareness multiplies the impact of individual expertise.

Section 9 | Closing Remarks

It has been a truly enriching experience to complete this Cyber Security Internship at Redynox. Working through these tasks has not only deepened my technical understanding of network security and web application vulnerabilities, but has also given me the confidence to approach real-world security challenges with a structured, methodical mindset.

I am grateful to the Redynox team for designing tasks that strike the right balance between conceptual learning and hands-on practice. Every task pushed me to think like both a defender and an attacker — a perspective that is essential in the ever-evolving field of cyber security. The opportunity to work with industry-standard tools such as Wireshark, OWASP ZAP, and WebGoat on my own hardware has been particularly valuable.

I leave this internship with a stronger foundation, a clearer career direction in cybersecurity, and a deep appreciation for the importance of proactive security practices. I sincerely thank the Redynox team for this opportunity and for their continued guidance throughout this journey. I look forward to applying these learnings in future projects and contributing meaningfully to the cybersecurity community.

Sai Sandeep Gedela

Cyber Security Intern

B.Tech CSE

June 2026

Redynox Team

Supervisor / Mentor

Redynox

June 2026

Section 10 | References

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- [5] **OWASP Top 10 — 2021** — <https://owasp.org/Top10/>
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- [8] **Redynox Company LinkedIn** — <https://www.linkedin.com/company/redynox/>